
RESEARCH PORTFOLIO

2023 – Present

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PRIMARY RESEARCH DOMAINS

Computational Intelligence & Adaptive Systems
Translational Engineering

01	MORPH-RL	Adaptive Morphing Wing Control via Deep RL	<i>Aeronautics</i>
02	IRIS	Cardiovascular Risk Detection via Retinal Imaging	<i>MedTech</i>
03	BIOLYZER	Breath-Based Biomarker Disease Detection	<i>Diagnostics</i>
04	AQUAKINETIC	Water-Based Kinetic Energy Harvesting	<i>Renewables</i>

01 | MORPH-RL

Adaptive Morphing Wing Control via Deep Reinforcement Learning

Aerospace Engineering | Autonomous Systems | MIT THINK Semifinalist 2025–2026

ABSTRACT: Small UAVs face a fundamental “endurance gap”—battery limitations restrict flight to 20–60 minutes. Fixed-wing designs compound this by operating efficiently only within narrow regimes. **MORPH-RL** is a novel adaptive control framework employing **Proximal Policy Optimization (PPO)** to govern morphing wing structures in real time. By commanding continuous, high-dimensional geometric deformations, the RL agent autonomously discovers configurations maximizing the lift-to-drag (L/D) ratio across variable flight conditions. Simulated trials show reward convergence at ≈ 500 epochs with a projected **10–15% energy improvement** over hand-tuned PID baselines—extending endurance for disaster response and environmental monitoring.

PROBLEM & PRIOR ART

UAVs in off-design conditions—wind shear, variable payload, altitude shifts—suffer surges in induced drag. Current morphing control systems (PID, MPC, fixed schedules) require manual tuning and optimize for *stability*, not *energy*. Key prior programs: DARPA Smart Wing (8–12% lift improvement), NASA ACTE (5–7% drag reduction), MIT MAD-CAT, and FishBAC ($>50\%$ L/D gain). **Critical gap:** no prior work applies deep RL to morphing control. MORPH-RL is the first system to *learn* optimal strategies from simulated experience, executing policy inference in <10 ms on embedded hardware.

DRL FRAMEWORK & MATHEMATICS

The system forms a closed loop: aircraft state $\rightarrow$ XFOIL aerodynamic lookup $\rightarrow$ neural network policy $\rightarrow$ morphing commands $\rightarrow$ JSBSim simulator $\rightarrow$ reward $\rightarrow$ PPO update.

Actor network outputs $\mathbf{u} = \mu(\mathbf{x}) + \sigma(\mathbf{x}) \cdot \varepsilon$ (camber, sweep, trailing-edge deflection). **Composite reward:**

$$R(t) = -\alpha P(t) - \beta |e_\alpha(t)| - \gamma |\Delta m(t)| + B(t)$$

$P(t)$: power consumption; $e_\alpha(t)$: pitch error; $\Delta m(t)$: morphing jerk penalty; $B(t)$: stability bonus. With $\alpha=1.0$, $\beta=0.1$, $\gamma=0.05$, the agent discovers energy trade-offs impossible to hand-code. JSBSim enforces physical constraints via $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, \mathbf{w})$. PPO stabilizes training:

$$\mathcal{L} = \mathbb{E} \left[\min \left(r_t \hat{A}_t, \text{clip}(r_t, 1-\varepsilon, 1+\varepsilon) \hat{A}_t \right) \right]$$

XFOIL precomputes C_L , C_D lookup tables enabling 1M simulated flights in 20–30 GPU hours.

METHODOLOGY & VALIDATION

Phase 1: JSBSim + XFOIL environment; baseline PID stable flight validated (± 5 m alt., ± 2 m/s speed). **Phase 2:** PPO pipeline; 1M steps, domain randomization (wind 0–10 m/s, altitude 50–500 m, payload $\pm 10\%$). **Phase 3:** 100+ test missions vs. PID baseline. **Phase 4:** OpenFOAM CFD validation; reward weight sensitivity analysis.

RESULTS & PERFORMANCE TARGETS

Emergent behaviors at convergence: agent reflexes trailing edge at cruise (reducing parasitic drag) and increases camber during loitering (preventing stalls). Targets:

Metric	Target
Energy improvement vs. PID	$\geq 10\%$
Pitch tracking error (RMS)	$< 2^\circ$
Speed deviation	$< 5\%$
Generalization degradation	$< 10\%$
Policy inference latency	< 10 ms

STATE & ACTION SPACE

Variable	D	Description
Airspeed V	1	m/s, continuous
AoA α	1	-15° to $+15^\circ$
Pitch rate q	1	deg/s
Altitude h	1	50–500 m
C_L , C_D	2	Aerodynamic coefficients
Wing geom. $\mathbf{g}$	6	Camber, sweep, TE deflection
Action $\mathbf{u}$	6	Δ shape commands

HARDWARE ROADMAP

- **Airfoil:** 3D-printed NACA 0012 with flexible TPU skin and carbon-fiber ribbing for continuous camber deformation.
- **Actuation:** Micro-servos at 8 spanwise stations commanding independent trailing-edge angles.
- **Controller:** ESP32 running quantized RL weights for real-time edge inference.
- **Validation:** Wind tunnel testing with force/torque sensor and PIV flow visualization.

SIMULATION-TO-REALITY & RISK MITIGATION

Unmodeled dynamics (actuator delay, sensor noise) are mitigated by injecting realistic delays during training and testing on perturbed aerodynamic models. Checkpoints saved every 50k steps guard against compute interruptions. Hard safety constraints (stall limits, deflection bounds) are enforced by JSBSim regardless of policy output, ensuring physically valid behavior throughout training.

ABSTRACT: CVD causes over 18 million deaths annually. Diagnostic access is limited in developing regions—professional fundus cameras cost \$25,000 and require trained specialists. **Iris** converts any smartphone into a clinical-grade retinal imaging system using a 3D-printed optical assembly and a deep learning pipeline. By measuring the **Arteriolar-to-Venular Ratio (avRatio)**, Iris identifies early-stage atherosclerosis and hypertension at a hardware cost of **\$26.50**, achieving **90% specificity** and enabling proactive CVD intervention **10+ years** before acute symptoms, published in **IEEE Xplore** following the MIT Undergraduate Research Technology Conference.

EXIGENCE & POVERTY CYCLE

The WHO estimates 550 million people suffer from CVD annually, excluding thousands undiagnosed due to financial and geographic barriers. CT angiography (\$1,500+) and contrast MRI (\$2,000+) use ionizing radiation and require specialists. CVD detected only after catastrophic events leaves families in medical debt, perpetuating intergenerational poverty. Iris breaks this cycle: \$26.50 per village versus \$2,000+ per person per year conventionally.

THEORETICAL BASIS: THE AVRATIO

The retina is the only location where the vascular system can be imaged non-invasively—microvascular changes mirror systemic circulatory disease:

$$\text{avRatio} = \frac{\sum D_a}{\sum D_v}$$

Healthy cohort: artery $135.63 \pm 15.64 \mu\text{m}$, vein $151.32 \pm 15.22 \mu\text{m}$, avRatio > 0.90 . An avRatio < 0.90 signals arterial narrowing from plaque buildup and venous dilation from hypertension, strongly correlating with coronary artery disease and stroke risk. Dr. Michael Blaha (Johns Hopkins) identifies avRatio measurements as “individualized, more concrete, and actionable” than other CVD screening methods.

OPTICAL HARDWARE (TIMS)

Iris uses a **Target Identifying Magnification System** consisting of a reusable 20D Indirect Ophthalmoscopy lens (\$26), 3D-printed biodegradable PLA alignment housing (\$0.50), and a synchronized smartphone flash with diffusion filter. Protocol: 3 images per eye (OD + OS), 6 total—all validated by TIMS alignment before submission. Compatible with any smartphone or standalone digital camera (offline upload supported).

AI PIPELINE & NEURAL ARCHITECTURE

(1) **Enhancement:** CLAHE + green-channel extraction for vessel contrast. (2) **Segmentation:** U-Net trained on DRIVE & STARE datasets for pixel-wise vascular mapping. (3) **Classification:** Arterioles vs. venules distinguished by color profile, tortuosity index, and branching geometry. (4) **Calibration:** Pixel distances converted to μm using known 20D lens magnification. (5) **Risk Scoring:** avRatio compared against ASCVD thresholds; patients at < 0.90 flagged for immediate cardiologist consultation.

COMPARATIVE PERFORMANCE

Feature	Traditional	Iris
Unit Cost	\$2k–\$25k	\$26.50
Portability	Heavy table-top	Pocket-sized
Analysis	Human expert	Automated ML
Power source	Wall outlet	Battery/mobile
Specificity	High	90%
Radiation	Yes (CT/MRI)	None
Training req.	Specialist	Minimal
Languages	English	126 languages

APP & VALIDATION

Iris app features encrypted patient profiles, offline AI inference (no internet required), 126-language support, and a physician portal for longitudinal avRatio trending. **Validated at 90% specificity** on smartphone-captured fundus images for hypertensive retinopathy; peer-reviewed and published in **IEEE Xplore** (MIT URTC).

FUTURE WORK & DEPLOYMENT ROADMAP

- **Near-term:** IRB-approved human participant trial comparing avRatio vs. clinical gold standards (echocardiogram, CAC score).
- **Mid-term:** 100+ unit deployment via Sub-Saharan Africa NGO partnerships; add glaucoma and diabetic retinopathy detection modules.
- **Long-term:** International market expansion; integrate CAC score for ASCVD event prediction; pursue FDA 510(k) clearance pathway.

03 | BIOLYZER

Breath-Based Biomarker Detection Platform for Type 1 Diabetes

Biosensing | Machine Learning | Non-Invasive Diagnostics | DRSEF 2024

ABSTRACT: Type 1 Diabetes demands relentless glucose monitoring, yet finger-prick tests are invasive and error-prone, leading to poor glycemic compliance. **BIOLyzer** non-invasively detects exhaled **acetone**—a Volatile Organic Compound (VOC) of ketosis—using a cross-reactive metal-oxide sensor array and an SVM classifier. With 93% of Type 1 patients exhibiting acetone >1.8 ppm, the biomarker provides a reliable metabolic surrogate. Prototype validation achieved **94.8% overall accuracy** and **97.33% accuracy for isopropyl acetone**, with mean measured concentration of 2.04 ± 0.21 ppm ($r^2 = 0.865$ correlation with glucose surrogates).

BIOCHEMICAL RATIONALE

In insulin-deficient individuals, β -oxidation produces acetone (C_3H_6O), which diffuses into exhaled breath. Healthy exhaled acetone: 0.3–0.8 ppm; ketotic patients: 1.8–5.0 ppm. Chemiresistive metal-oxide sensors quantify sub-ppm shifts via conductivity changes at $300^\circ C$. NIH confirms 93% of Type 1 patients exhibit elevated acetone; WHO reports 1.5 million annual diabetes deaths, 100,000 undiagnosed.

SENSOR ARRAY & HARDWARE

Multi-sensor “Electronic Nose” configuration:

- **MQ-138 VOC:** Primary acetone detection, sub-ppm sensitivity.
- **MQ-135:** Cross-validation for ammonia, alcohol, and acetone.
- **Pre-heating:** 5V heating elements at $300^\circ C$ for stable gas adsorption.
- **Baffle Chamber:** 3D-printed (Fusion 360) PLA enclosure directing uniform airflow while minimizing condensation—the primary failure mode in breath analysis.
- **Display:** 1.3” IIC OLED LCD for real-time PPM read-out.

SIGNAL PROCESSING & ML

Raw voltage from Arduino 10-bit ADC is baseline-corrected:

$$\Delta V = V_{\text{peak}} - V_{\text{ambient}}$$

A **Support Vector Machine (SVM)** classifier trained on controlled acetone dilutions (1.5–3.5 ppm; formula: mg/L) distinguishes diabetic acetone spikes from background noise. Decision thresholds: >1.8 ppm $\rightarrow$ risk warning; ≤ 0.8 ppm $\rightarrow$ healthy; intermediate $\rightarrow$ monitor.

EXPERIMENTAL VALIDATION

Condition	Conc. (ppm)	Accuracy
Isopropyl acetone	Various	97.33%
Healthy volunteers (avg.)	0.42 ± 0.11	—
Diabetic simulation (avg.)	2.04 ± 0.21	94.8%

Strong positive correlation confirmed: $r^2 = 0.865$ vs. PubMed reference datasets.

ERROR ANALYSIS & LESSONS LEARNED

Error Source	Mitigation
Breadboard wiring	Color-coded layout; continuity verified before power-on
5V/GND rail confusion	Labeled rails; circuit diagram reviewed each session
Sensor drift	24-hour burn-in at room temp before calibration
Condensation	PLA baffle redesigned for laminar airflow
Acetone evaporation	Sealed Erlenmeyer flask; capped immediately after prep

MOBILE APP & PHYSICIAN INTEGRATION

Companion app enables longitudinal acetone tracking, weekly trend charts, and a physician portal for remote insulin regimen adjustment via breath trends. Profile-based tracking allows patients to monitor acetone averages over time and flag anomalous readings for clinical follow-up.

ROADMAP & NEXT STEPS

By swapping sensor profiles and retraining the SVM, BIOLyzer expands to: oncology (isoprene/benzene for lung cancer), hepatology (ammonia for cirrhosis), nephrology (dimethylamine for kidney disease), and environmental monitoring (CO , NO_2). An IRB-approved study will compare readings against CGMs to establish a breath-ppm to blood-mg/dL conversion factor, transitioning BIOLyzer to a quantifiable diagnostic. Hardware v2 targets a miniaturized MEMS sensor (sub-ppb) with Bluetooth connectivity.

04 | AQUAKINETIC

Water-Based Kinetic Energy Harvesting from Slow-Moving Ocean Currents

Renewable Energy | Marine Engineering | DRSEF 2025

ABSTRACT: Wind and solar energy suffer from intermittency requiring expensive storage. Slow-moving water currents (<2.0 m/s) are **consistent and fully predictable**, yet inaccessible to conventional hydroelectric systems requiring high head or velocity. **AquaKinetic** targets this untapped market with a buoy-driven cable-and-drum transduction system and a permanent-magnet generator. Testing at 1.0, 1.5, and 2.0 m/s demonstrated stable voltage output across all 5 trials per speed, achieving peak mechanical power of **51.49 W** and **89% system efficiency at 1.5 m/s**—demonstrating strong scalability for remote coastal communities and autonomous oceanic sensor networks.

HYDROKINETIC OPPORTUNITY & CONTEXT

The IPCC mandates 45% emissions reduction by 2030 and net-zero by 2050. Marine kinetic energy is non-intermittent, requires no land use, and has minimal visual impact. Global slow-moving currents (<1.5 m/s), mapped in NASA OSCAR ocean current data, represent a massive untapped resource. Existing competitors (CorPower Wave Spring, IHI Single Turbine, SIT Buoy Spring) require higher velocities or prohibitive capital investment. AquaKinetic targets the 80% of water energy sites excluded by conventional turbines.

MECHANICAL DESIGN & TRANSDUCTION

Multi-stage energy conversion chain: **(1) Buoy** captures current-driven oscillation, driving a retracting cable—neutral buoyancy means no heavy foundation. **(2) Cable & drum** converts linear motion to rotational torque with minimal tension loss. **(3) Permanent-magnet generator** (60–120 RPM) produces stable output at low speeds. **(4) Waterproof GFRP casing** with heat-dissipation ventilation ensures >20-year lifespan.

POWER CONDITIONING

Raw 3-phase AC is processed through: full-wave **rectification** (AC → DC), **MPPT logic** (dynamic impedance adjustment for maximum energy extraction at all velocities), and **voltage regulation** (clean 5V/12V DC output for Li-Ion charging or IoT sensor power).

EQUATIONAL ANALYSIS

Mechanical power $P_{\text{mech}} = \tau\omega$:

$$1.0 \text{ m/s: } (2.35)(10.3) = 24.29 \text{ W}$$

$$1.5 \text{ m/s: } (3.30)(15.5) = 51.49 \text{ W}$$

$$2.0 \text{ m/s: } (3.43)(12.1) = 41.50 \text{ W}$$

Theoretical output $P = \frac{1}{2}\rho Av^3 C_p$:

$$1.0 \text{ m/s: } 13.75 \text{ W } (C_p = 68.2\%)$$

$$1.5 \text{ m/s: } 25.82 \text{ W } (C_p = 85.4\%)$$

$$2.0 \text{ m/s: } 16.53 \text{ W } (C_p = 41.0\%)$$

Energy stored $E = Vit$:

$$1.0 \text{ m/s: } (24)(1.01)(1 \text{ h}) = 24.27 \text{ Wh}$$

$$1.5 \text{ m/s: } (24)(2.14)(1 \text{ h}) = 51.48 \text{ Wh}$$

$$2.0 \text{ m/s: } (24)(1.56)(1 \text{ h}) = 37.44 \text{ Wh}$$

EXPERIMENTAL RESULTS

Spd	Trial	RPM	V	A	W	Eff.
1.0	1	100	10.5	2.1	20.1	61%
	2	101	11.5	2.1	21.1	71%
	3	97	11.1	2.2	20.5	68%
	4	95	10.7	2.3	22.0	76%
	5	99	10.9	2.4	20.9	65%
1.5	1	146	14.7	3.1	46.1	81%
	2	151	15.6	3.4	45.3	89%
	3	149	15.4	3.6	46.2	86%
	4	150	15.1	3.2	45.9	87%
	5	147	14.9	3.7	45.3	84%
2.0	1	121	12.1	2.6	32.1	36%
	2	132	13.1	2.8	31.8	42%
	3	126	12.4	2.4	31.5	47%
	4	128	12.6	2.1	30.9	41%
	5	126	12.8	2.4	31.1	39%

Peak efficiency bolded per condition. Efficiency drop at 2.0 m/s reflects mechanical drag at elevated RPM—confirmed by stable voltage repeatability ($\sigma < 0.5 \text{ V}$).

SUSTAINABILITY & ECOLOGY

AquaKinetic is fish-friendly: slow rotation and absence of sharp edges prevent aquatic harm. Neutral buoyancy enables “plug-and-play” deployment without construction equipment—critical for remote island and coastal communities. GFRP corrosion resistance ensures a >20-year service life with minimal maintenance.

FUTURE DIRECTIONS

- **CFD:** Advanced blade geometry to lower threshold to <0.5 m/s.
- **Green Hydrogen:** Surplus energy to on-site electrolysis for zero-emission storage.
- **Remote Electrification:** Base-load power for off-grid island communities.
- **Marine Research:** Autonomous sensor power for ocean acidification monitoring.